magic-cubes

v0.0.1 MIT

Represent Rubik's cubes of arbitrary size and apply algorithms to them

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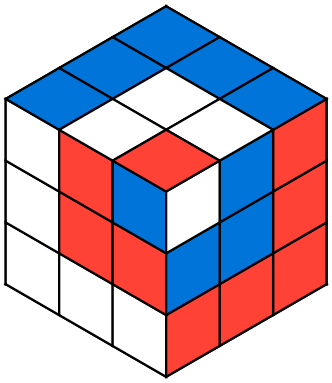
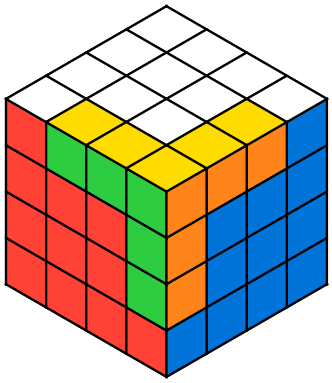


U' L' U' F' R2 B' R
F U B2 U B' L U' F
U R F'



F U2 L F L' B L U
B' R' L' U R' D' F'
B R2

Table of Contents

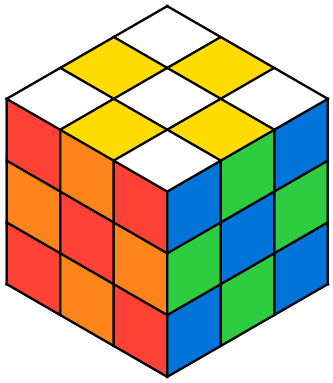
Part I

About

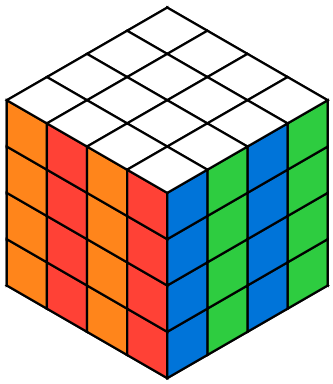
In 1974, Ernő Rubik invented a mechanical puzzle and called it the *Magic Cube*. Years later, in 1980, the puzzle was renamed the *Rubik's Cube*, name by which it is now known. Today, it is considered to be the world's bestselling puzzle game.

magic-cubes is a package built on top of CeTZ that allows you to create, manipulate and render Rubik's cubes of any size.

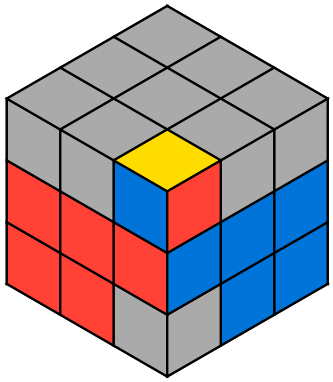
```
#draw_cube(
  apply(
    cube(),
    "F2 B2 R2 L2 U2 D2"
  )
)
```



```
#draw_cube(
  apply(
    cube(size: 4),
    "R2 3R2 F2 3F2 R2 3R2"
  )
)
```



```
#draw_cube(
  apply(
    inverted: true,
    f2l-cube,
    "R2 U R2 U R2 U2 R2"
  )
)
```



Part II
Guide

II.1 Creating cubes

Cubes can be created using the “cube” function…

II.2 Applying algorithms

II.2.1 Moves

The following moves are supported, more will be added in future updates:

- F: front face
- R: right face
- U: up face
- B: back face
- L: left face
- D: down face

All of them represent a single **clockwise** turn in the respective face.

II.2.2 Modifiers

Adding an apostrophe (') or a “2” after the move will cause an anticlockwise turn or a double one (180º) respectively.

II.3 Rendering cubes

Part III

Full reference

III.1 Types

(
  ⟨f⟩ dictionary
  ⟨r⟩ dictionary
)

III.2 Functions

#apply(⟨cube⟩, ⟨alg⟩, ⟨inverted⟩: false) → **cube-t**

Returns a new cube after applying the algorithm.

Argument

⟨cube⟩ cube-t

The cube to apply the algorithm.

Argument

⟨alg⟩ str

The algorithm to apply.

Argument

⟨inverted⟩: false bool

Whether it should be applied in inverse order.

#draw_cube(⟨cube⟩, ⟨x⟩: 35.264deg, ⟨y⟩: 45deg, ⟨z⟩: 0deg)

Draws a cube

Argument

⟨cube⟩ cube-t

the cube to draw.

Argument

⟨x⟩: 35.264deg angle

the x angle.

Argument

⟨y⟩: 45deg angle

the y angle.

Argument

⟨z⟩: 0deg `angle`

the z angle.

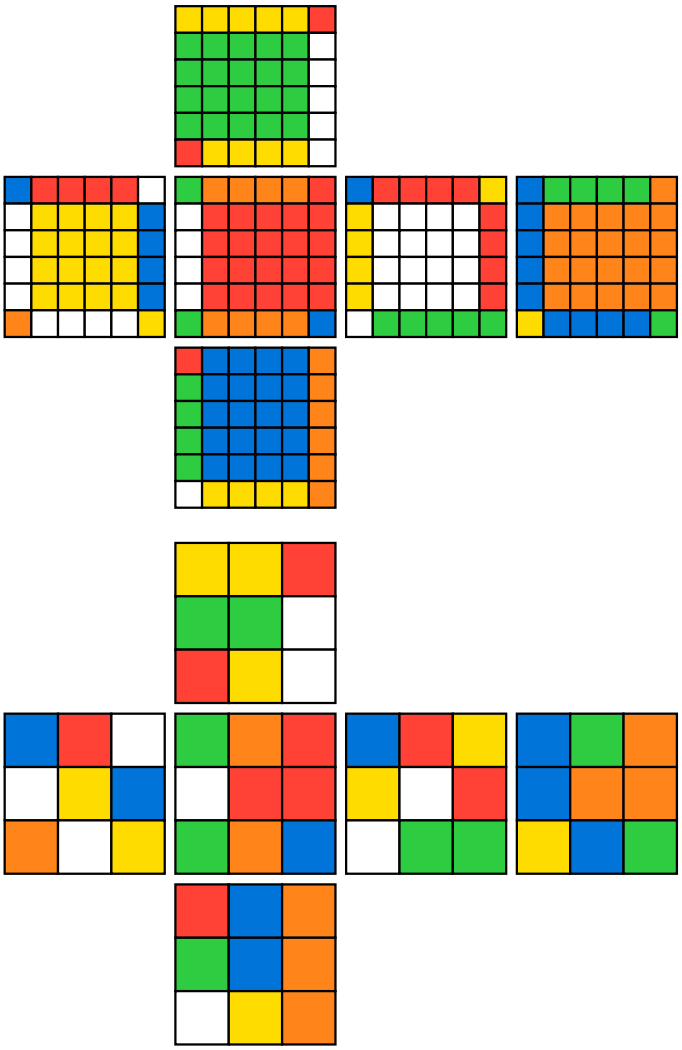
#draw_flat(⟨cube⟩)

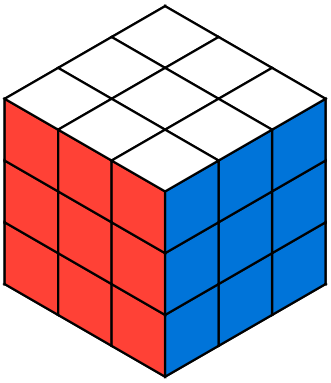
Draws a flat cube.

Argument

⟨cube⟩ `cube-t`

The cube to draw.





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Part IV

Index

A

C

D